RECLAIMING THE EARTH WE FARM:
SUSTAINABLE AGRICULTURE AND ENVIRONMENTAL ACTION AS A GLOBAL STRATEGY AGAINST ANTIMICROBIAL RESISTANCE

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1. Introduction

"The Citarum River used to be so beautiful back then," said Ibu Edah, a resident interviewed recently. She has lived along its banks since 1968 and witnessed firsthand how dramatically the river has changed. Before the year 2000, the water was remarkably clean, clean enough to bathe in, wash clothes, and use for other daily needs without a second thought. Ibu Edah does not know the worst part. She does not know that the water she uses contains antibiotic residues that are silently conditioning bacteria to become untreatable.

Today, the river is streaked with textile dye, thick with sewage, and choked with garbage. Yet the most dangerous threat it carries cannot be seen, smelled, or tasted. Ibu Edah's deepest hope is simple: that one day the river will be free from industrial waste, and that people will stop treating it as a dumping ground.

Antibiotic residues in water are conditioning bacteria to become untreatable. Continuously exposed to low concentrations of antibiotics from farms, hospitals, and factories upstream, these bacteria develop resistance to the very medicines on which modern healthcare depends.

This is not an isolated case; similar patterns occur globally from the Ganges river to the yellow river to the Mississippi river. This turns the world's waterway into a place where drug-resistant bacteria form. This phenomenon, antimicrobial resistance (AMR), has been identified by the WHO as one of the ten greatest threats to global health. When researchers from the National Research and Innovation Agency (BRIN) detected amoxicillin residues in every Upper Citarum water sample tested in 2021, they were describing the watershed, the food chain, and the future of 28 million people.

This essay identifies agricultural antibiotic pollution as the main driver of antimicrobial resistance (AMR) in rivers, tracing its pathway from farm to human infection under the One Health framework. It argues that an effective response requires combining sustainable farming practices (composting mandates, biochar filtration, and bans on growth-promoting antibiotics) with advanced wastewater treatment, while highlighting the critical role of future leaders in policy implementation.

2. The Problem: A River, a Resistance, and a Global Emergency

2.1 The Citarum River

The Citarum River stretches approximately 300 kilometres through the heart of West Java, supplying drinking water to 80% of Jakarta's populations, irrigating over 400,000 hectares of rice paddies, and sustaining the livelihoods of fishing and farming communities across the region. It is the biological and economic foundation of one of the most densely populated areas on Earth. However, until recently, it is widely recognised as one of the most polluted rivers in the world, receiving over 2,000 factories' industrial discharge, around 280,000 tons of untreated human sewage annually, agricultural runoff containing antibiotics and pesticides, and approximately 20,000 tons of plastic waste per year (ADB, 2016; Greenpeace, 2019; Yudo & Said, 2018). In some sections, dissolved oxygen levels are so low that the Indonesian government has declared the river biologically dead.

2.2 Understanding AMR: From the River to the Bedside

To understand what happens when resistant bacteria enter the human body, consider the story of Pak Ahmad, a rice farmer cuts his foot on a piece of rusty metal. The wound becomes infected. The doctor prescribes amoxicillin. Nothing happens. By the time he reaches a hospital in Bandung, the infection has spread to his bones. Antibiotic after antibiotic is tried. Nothing works. Pak Ahmad dies two weeks later from a wound that would have been resolved by penicillin in a previous generation. The bacteria responsible were resistant to every available treatment, shaped by the very river where he had farmed and bathed for decades.

Antibiotics function as molecular keys designed to open specific bacterial locks. When bacteria are repeatedly exposed to sublethal antibiotic concentrations, they mutate those locks until no existing key fits. The molecular mechanism is well documented: bacteria acquire resistance through gene mutations or by absorbing resistance genes from neighbouring organisms via horizontal gene transfer (HGT). In Citarum sediments, researchers have detected blaCTX-M, conferring resistance to broad-spectrum cephalosporins; mcr-1, conferring resistance to last resort colistin and tetracycline resistance genes tetA, tetB, and tetM. These genes do not remain in the river; they travel in water droplets, on food surfaces, and through the hands of children into households and human bodies (Larsson & Flach, 2022).

2.3 Global Stakes

The scale of the AMR crisis is staggering. It currently causes 1.27 million deaths directly and is associated with 4.95 million deaths annually, equivalent to more than 3,500 lives every single day. By 2050, projections place AMR-related deaths at 10 million per year, surpassing current cancer mortality, with an estimated $100 trillion in economic costs (Murray et al., 2022; WHO, 2019; World Bank, 2017). Routine surgical procedures, chemotherapy, and the treatment of common infections could all become life-threatening as resistance advances. This demonstrates that AMR is not a distant scientific concern, but an immediate global crisis with profound human and economic consequences.

3. The Citarum as an Incubator of Resistance

3.1 The Three Sources of Antibiotic Pollution

Antibiotic contamination in the Citarum originates from three interconnected sources: pharmaceutical factories, which can release enough antibiotics to treat 45,000 people daily (Lubbert et al., 2017); hospitals and clinics, through which up to 70% of each antibiotic dose is excreted unchanged via untreated sewage; and agriculture, which is globally the largest and most overlooked source, responsible for 73% of all antibiotic consumption worldwide (Van Boeckel et al., 2019).

3.2 Agriculture: The Primary Driver

Farmers administer antibiotics to poultry, pig, and fish feed to accelerate growth and to prevent disease in densely crowded factory-farm conditions. Between 50% and 70% of each dose is excreted unchanged in animal waste, still biologically active. When farmers spread this raw manure onto rice paddies as fertilizer, rainfall carries these active antibiotics and resistant bacteria into local streams that eventually drain into the Citarum. As a result, bacterial communities are continuously exposed to sub-lethal concentrations that do not eliminate them but instead exert selective pressure, progressively favoring resistant strains. In 2021, Indonesia's National Research and Innovation Agency (BRIN) detected amoxicillin residues in every water sample collected from the Upper Citarum. In addition, ciprofloxacin has been discovered at up to 50 micrograms per litre in South Asian rivers, roughly 500 times above the safe resistance-selection threshold (Larsson & Flach, 2022). Similarly, tetracycline levels in West Java livestock wastewater reached between 140 and 380 micrograms per litre, concentrations sufficient to actively drive resistance rather than merely indicate contamination (Manyi-Loh et al., 2018).

At the genetic level, the presence of the mcr-1 gene has been identified in *Escherichia coli* from Citarum-adjacent poultry farms, and resistance markers have also been detected in river sediment at up to 10^8 gene copies per gram (Bengtsson-Palme et al., 2018). Taken together, this evidence confirms that agricultural practices are not peripheral, but central to the emergence and spread of antimicrobial resistance.

4. The AMR Escalation Cycle and Environmental Consequences

AMR does not produce a constant level of risk, it drives a self-reinforcing escalation. Antibiotic emission selects for resistant bacteria, which cause treatment failures, which force doctors to use stronger last-resort drugs, which are more environmentally persistent, which select for resistance to those drugs, until no therapeutic options remain. Alongside this medical escalation, the environmental consequences are compounding: resistant bacteria degrade soil fertility, raise water treatment costs, accumulate in fish tissue, enter crop plants through contaminated irrigation water, and kill the natural microbial communities that give rivers their self-cleaning capacity. Most critically, unlike plastic or petroleum pollution, genetic pollution is irreversible on human timescales. Antibiotic resistance genes can persist in sediment for decades or centuries (Bengtsson-Palme et al., 2018). Even if all antibiotic use ceased today, future bacteria could inherit resistance from ancestors that never encountered a single antibiotic molecule. This permanence makes prevention categorically superior to remediation and renders early, systematic action the only practically defensible response. This self-reinforcing cycle underscores the urgency of intervention before resistance reaches irreversible levels

5. Solutions: An Integrated One Health Strategy

5.1 The One Health Framework

The WHO, FAO, and the World Organisation for Animal Health (OIE) all endorse the One Health framework: the recognition that human health, animal health, and environmental health form a single, inseparable system. Because AMR originates at the intersection of all three domains, no single-sector intervention is sufficient. The solutions below are designed to address agricultural practice, environmental contamination, and human health simultaneously.

5.2 Addressing Common Counterarguments

On food security: When the European Union banned growth-promoting antibiotics in 2006, livestock antibiotic use fell by 50% without a collapse in meat production (European Medicines Agency, 2017). Danish pig production actually increased after the ban, as farmers adopted improved hygiene and vaccination. Sustainable alternatives exist and demonstrably work.

On cost: AMR, if unchecked, could push 28 million people into pwsoverty by 2050 (World Bank, 2017). Biochar production from rice husks costs approximately $0.05 per litre of treated effluent. Government subsidies for composting facilities and farmer training are investments in public health infrastructure, not expenditures to be avoided.

5.3 Evidence-Based Policy Solutions

Six coordinated interventions constitute a comprehensive policy response. First, Indonesia must enforce Permentan No. 14/2017, which bans antibiotic growth promoters in feed, through regular inspections, meaningful penalties, and transition subsidies for smallholders. Second, manure composting for a minimum of 60 days before field application must be mandated; this destroys 90% to 99% of resistant bacteria and can be delivered through subsidised community facilities (Manyi-Loh et al., 2018). Third, biochar filters from locally abundant rice husks should be deployed at farm level; a 2022 study confirmed 94% removal of *Pseudomonas aeruginosa* and 88% removal of clarithromycin (Xiang et al., 2022). Fourth, farmer training in hygiene, vaccination, and composting reduced antibiotic use by 40% in Thai poultry farms within two years and is directly replicable in Indonesia (FAO, 2019). Fifth, advanced wastewater treatment, including ozonation (99% bacterial removal), activated carbon filtration (90% to 95% antibiotic removal), and membrane bioreactors (99.9% bacterial removal), must be mandatory for all Citarum-basin factories and hospitals (Sundin et al., 2021). Sixth, bacteriophage therapy and probiotic supplementation, which reduced Salmonella infections in poultry by 70%, can replace antibiotics without generating resistance (Callaway et al., 2018; Kutter et al., 2019). These solutions demonstrate that the AMR crisis is not only preventable, but already equipped with viable, evidence-based interventions.

5.4 The Role of Future Leaders

Systemic crises of this magnitude are ultimately resolved by the generation that inherits them. As advocates, young leaders can demand enforcement of existing legislation such as Permanent No. 14/2017, organise evidence-based public campaigns, and hold governments and corporations accountable. Indonesian students, living within the Citarum watershed and bearing the direct health consequences of AMR, possess particular moral standing and credibility to do so. As innovators, students in agriculture, environmental science, and engineering can develop locally adapted solutions including low-cost biochar filter designs for smallholder farms and mobile systems for monitoring antibiotic residues in local waterways. As bridge-builders, future leaders can close the gap between scientific evidence and policy action by translating complex AMR data into accessible narratives that motivate change across sectors and communities. That communication gap is, at its core, a leadership gap, and it is precisely the space that the next generation is best positioned to fill.

5.5 Individual Actions That Matter

Policy reform and individual behaviour are mutually reinforcing, not competing strategies. The following evidence-based actions create the market signals and social norms that enable systemic change: never purchase antibiotics without a prescription; always complete the full prescribed course; ask your doctor whether an antibiotic is genuinely necessary; choose antibiotic-free meat and eggs to reward sustainable farming; support local producers and farmers markets and share knowledge about AMR, since awareness is the indispensable first step toward both policy change and personal behaviour change.

6. Conclusion

This essay has demonstrated that agricultural antibiotic pollution is one of the primary drivers of AMR in river ecosystems such as the Citarum, operating through a well documented chain from livestock farms to river contamination to human infection and death. The evidence is unambiguous: resistance genes, including mcr-1, blaCTX-M, and tetracycline markers are present at concentrations far exceeding safe thresholds, and the genetic pollution they represent is, on human timescales, irreversible.

The six solutions presented here are not theoretical aspirations. They are evidence-based interventions with documented effectiveness across multiple national contexts, from the European Union to Thailand to Georgia. They are united by the One Health principle that agricultural practice, environmental integrity, and human health form a single system, and that protecting one requires protecting all three.

The political will and implementation capacity required to put these solutions into practice will come from the generation now in school and university. Indonesian students, living within the Citarum watershed and equipped with scientific understanding, are not bystanders to this crisis. They are its most credible advocates, its most adaptable innovators, and its most essential bridge builders between evidence and action. We all know that the river has changed and that the medicines no longer work as they should. We don't know the terminology, but we know the consequences. That consequence is preventable. The knowledge exists. The solutions are proven. The remaining question is whether the next generation of leaders will act before the window for prevention closes entirely.

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An optional interactive version is available online for supplementary exploration. Featuring visual data explorations, an interactive AMR escalation cycle, and photographic documentation of the Citarum River, is available at:

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